SCOPE : Malkuth — Reality

senkaL8

Chapter 0. Introduction

The Sephirotic names used in this paper (such as Malkuth, Yesod, and Keter) are not used in their religious or mystical sense. Within SCOPE, they are used solely as labels for identifying cognitive structures.

SCOPE (Sephirotic Cognitive Process Engine) is a coordinate system for estimating the cognitive structures of both humans and AI.

In this context, a coordinate system does not represent a flowchart in which cognitive processing proceeds through a fixed sequence. Nor does it assume that cognition begins from any particular layer.

A single cognitive process is not necessarily associated with only one layer. Depending on the situation, it may correspond to multiple layers, and the same correspondence may appear repeatedly. Furthermore, the way these correspondences emerge may vary according to the individual and the situation.

For this reason, SCOPE does not define a fixed processing order such as "cognition begins at L0" or "L8 is the final destination." Each layer is not intended to represent hierarchy or temporal sequence, but rather serves as a coordinate for estimating and organizing cognitive structures.

This series begins with Malkuth as the entry point to SCOPE. Within the outputs included in Malkuth, this series primarily focuses on language-based outputs such as conversations and written text. The applicability of SCOPE to other forms of output remains a subject for future investigation.

Chapter 1: What Is Malkuth

Malkuth is the layer at which the result of cognitive processing is output into reality and becomes observable.

Here, "reality" refers to the physical world in which humans live. What matters is not that cognitive processing remains confined within the mind, but that it comes to exist as part of reality by being externally expressed.

"Observable" does not mean that someone has actually observed it. It means that the result of cognitive processing has been output into reality and has reached a state in which it can be observed, including by the originator.

Speech, writing, facial expressions, physical movement, music, painting, dance, and program code, for example, are all instances of cognitive processing whose results have been output into reality, and are therefore included in Malkuth.

By contrast, thought that remains solely within the mind, or cognitive processing that has not yet been externalized, is not included in Malkuth.

What matters here is not the correctness or value of what is output. Whether it is true or false, whether it is art or idle talk, the moment the result of cognitive processing is output into reality and becomes observable, it is treated as Malkuth.

SCOPE takes Malkuth as its object of observation and, from it, estimates which cognitive structures may have been at work behind it.

Malkuth is not defined as the starting point or the endpoint of cognitive processing. It is positioned as an observable point of contact at which the result of cognitive processing appears in reality. What is output into Malkuth can become the occasion for new cognitive processing, which may again be output into Malkuth.

Chapter 2. A Layer Shared by Humans and AI

In Chapter 1, Malkuth was defined as the layer in which the results of cognitive processing are expressed in reality and become observable.

This definition is not intended to apply only to humans. In AI as well, the internal processing itself cannot be directly observed. However, the results of that processing, such as generated text and responses, appear in an observable form.

For humans, outputs appearing in Malkuth include speech, conversations, self-talk, written text, diaries, chats, and social media posts. All of these are instances in which internal cognitive processing appears in an observable form through language.

In AI, generated text, conversational responses, summaries, translations, and code also appear as outputs. These are not identical to human cognitive processing. However, they correspond to human linguistic outputs in that they represent the results of internal processing appearing in an observable form.

Therefore, SCOPE does not treat humans and AI as the same. Their internal structures are different. However, in both cases, SCOPE does not rely on the internal processing itself, but rather on the observable results that appear as outputs. In this sense, Malkuth may serve as a common point of observational contact from which both can potentially be compared and their underlying cognitive structures estimated.

When mapped to an LLM, Malkuth corresponds to the output layer. This does not imply that the human Malkuth and the LLM output layer perform the same processing. What corresponds between them is the structural role in which the results of internal processing appear as linguistic outputs and become observable.

Thus, Malkuth is not a concept for treating humans and AI as the same. Rather, it is the layer that provides a common point of contact through observable outputs for examining subjects with different internal structures.

Chapter 3: Malkuth Alone Does Not Reveal the Interior

Malkuth is the layer at which the result of cognitive processing appears in an observable form. However, what can be observed is only the output itself; it does not directly reveal the cognitive processing that occurred behind it.

Even when the output is identical, the underlying internal state is not necessarily the same.

For example, the Japanese expression "daijoubu desu" (大丈夫です) — often rendered in English as "I'm fine," "It's okay," or "No, thank you" — can arise from genuinely feeling fine, from concealing anxiety, from hesitation or reserve, or from resignation. The same surface expression may originate from markedly different cognitive states. The fact that this single expression splits into several distinct English translations is itself evidence of the ambiguity this chapter is pointing to: one observable output does not correspond to one determinate internal state.

The same structure holds for AI. Even if internal numerical information could be examined, it remains difficult to determine with certainty which internal processing gave rise to a given output.

In this way, what can be observed from Malkuth is the output itself, and the internal state cannot be uniquely determined from that output alone.

Therefore, observable output alone cannot reveal cognitive structure directly. What is needed is a framework for organizing and estimating, from the observed result as a clue, which cognitive structures may have been at work.

SCOPE is positioned as the coordinate system that performs this estimation. By treating Malkuth as the point of observational contact and estimating the cognitive structure behind it, SCOPE organizes cognitive processing that cannot be captured by observable output alone.

Chapter 4. Connection with COS

Chapter 3 showed that Malkuth allows us to observe only the output itself and does not enable us to determine the underlying cognitive structure uniquely. Therefore, SCOPE estimates cognitive structures by using the observed output as its point of reference.

In contrast, COS (Cognitive Operating System) is a framework for organizing cognitive processes in humans and AI.

For example, when observing a single utterance, COS organizes the cognitive processes leading to that utterance from the perspectives of decision-making, interpretation, transmission, and observation. SCOPE, on the other hand, uses that utterance as Malkuth's point of observational contact and estimates which cognitive structures may have been involved.

Thus, although COS and SCOPE deal with the same output, they focus on different aspects of it. COS organizes cognitive processes, whereas SCOPE estimates cognitive structures. They are not interchangeable but are independent models that approach the same Malkuth from different perspectives.

Chapter 5. Malkuth as the Starting Point of Observation

In this paper, Malkuth has been defined as the layer where the results of cognitive processing appear in an observable form.

It is important to emphasize that Malkuth is neither the beginning nor the end of cognitive processing.

SCOPE does not assume that cognition proceeds in a fixed sequence. Therefore, it does not define processing orders such as "cognition begins from Malkuth" or "cognition ultimately reaches Malkuth."

However, inferring cognitive structures requires some form of observable result.

For humans, this may be speech or written text; for AI, it may be generated responses. Once the results of cognitive processing appear in an observable form, it becomes possible to infer the underlying cognitive structure.

In this sense, Malkuth is positioned not as the starting point of cognition, but as the point of observational contact. In this paper, observable outputs are treated as the point of observational contact from which cognitive structures are inferred.

Accordingly, Malkuth is introduced first in this paper not because cognition begins there, but because it serves as the entry point for examining cognitive structures through observable outputs.

The next paper will examine L0, the cognitive layer related to survival and bodily responses.

Conclusion

This paper has presented Malkuth as the layer where the results of cognitive processing become observable.

Malkuth is neither the beginning nor the end of cognitive processing. Nor does it represent cognitive structure itself.

In SCOPE, observable outputs serve as the point of observational contact from which underlying cognitive structures are inferred. For this reason, Malkuth has been introduced first in this paper as the entry point to the SCOPE framework.

What has been presented in this paper is the starting point for examining cognitive structures through observable outputs.

The next paper will focus on Layer 0 (L0), the cognitive layer related to survival and bodily responses. It will examine the cognitive structures associated with survival and bodily reactions.